



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- *Summary of methodologies*
- *Summary of all results*

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Introduction

Project background and context

SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.

Problem

- Predict if the Falcon 9 first stage will land successfully.

Section 1

Methodology

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Methodology

Executive Summary

- Data collection methodology:
 - Data was collected from an API (SPACEX API)
- Perform data wrangling
 - Number of launches on each site, number and occurrence of each orbit, number and occurrence of mission outcome of the orbits, and Creating a landing outcome label from Outcome column.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

Data sets were collected using

- SpaceX API
- Web Scraping

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Data Collection – SpaceX API

Data Collection Process:

- Requested core launch data from the SpaceX API launches using Python's requests library.
- Filtered dataset to isolate Falcon 9 launches by unpacking nested JSON structures.
- Utilized targeted API functions (getRocket, getLaunchSite, getPayloadData, getCoreData) to convert entity IDs into descriptive features like payload mass, orbit type, and landing success.
- **Github url:**
<https://github.com/ibrahimlois58-sketch/project-1/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Fetch Data: HTTP GET request to SpaceX past launches endpoint.

Filter: Unpack JSON data and isolate the 90 Falcon 9 launch rows.

Map IDs: Run helper functions (getRocket, getPayloadData, etc.) to resolve API ID strings into actual names.

Clean: Handle 5 missing payload mass by calculating and replacing them with the column mean.

Export: Save the completed 90x17 dataframe to a CSV file.

Data Collection - Scraping

Data Collection Process:

- Performed an HTTP GET request to retrieve the Wikipedia page for Falcon 9 launch records using the requests library.
- Parsed the raw HTML content with BeautifulSoup to isolate all structured launch tables.
- Extracted column headers by filtering out text and HTML tags, keeping only valid, non-empty text elements.
- Iterated through the HTML table rows (<tr> and <td>), applying custom parsing functions to isolate launch dates, times, booster versions, launch sites, payloads masses, orbits, customers, and flight outcomes.

GitHub URL:

- [https://github.com/ibrahimlois58-sketch/project-1/blob/main/jupyter-labs-webscraping%20\(1\).ipynb](https://github.com/ibrahimlois58-sketch/project-1/blob/main/jupyter-labs-webscraping%20(1).ipynb)

Target URL: Wikipedia Falcon 9 Launch List

Fetch raw HTML webpage using requests.get()

Create BeautifulSoup object; isolate 'wikitable' elements containing launch records

Loop through table rows (<tr>) and cells (<td>) for row by row data extraction

Append extracted rows into a master launch_dict, Convert populated dictionary into Pandas DataFrame

Data Wrangling

Data Wrangling Process:

- Analyzed launch data to determine the frequency and distribution of launches across key sites (e.g., CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E).
- Determined landing outcomes using the Outcome column and categorized them into binary success (1) or failure (0) metrics.
- Engineered a target column named Class using a conditional loop to serve as the ground truth label for downstream predictive modeling.
- Profiled orbit types (LEO, GTO, ISS, PO, etc.) to assess their respective launch success rates.

GitHub URL:

- <https://github.com/ibrahimlois58-sketch/project-1/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

charts Summary

- Catplot to determine relationship between Flight Number and Launch Site
- Scartter plot to determine the relationship between Payload Mass and Launch Site, Flight Number and Orbit type, and Payload Mass and Orbit type
- Barchart to determine the relationship between the Success Rate of each Orbit type
- Line chart to Visualize Launch Success yearly trend

GitHub URL:

<https://github.com/ibrahimlois58-sketch/project-1/blob/main/edadataviz.ipynb>

EDA with SQL

Summary of SQL queries performed

- `DROP TABLE IF EXISTS SPACEXTABLE;`
- `CREATE TABLE SPACEXTABLE AS SELECT * FROM SPACEXTBL WHERE Date IS NOT NULL;`
- `SELECT name FROM sqlite_master WHERE type='table';`
- `SELECT DISTINCT launch_site FROM SPACEXTABLE;`
- `SELECT Booster_Version FROM SPACEXTABLE`
- `WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);`
- `SELECT Landing_Outcome, COUNT(*) AS Outcome_Count`
- `FROM SPACEXTABLE`
- `WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'`
- `GROUP BY Landing_Outcome`
- `ORDER BY Outcome_Count DESC;`

GitHub URL:

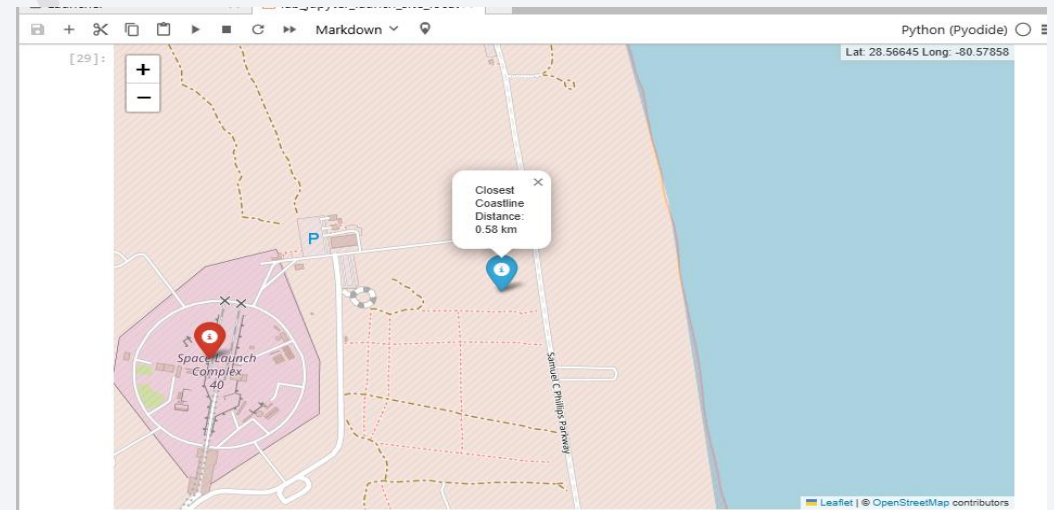
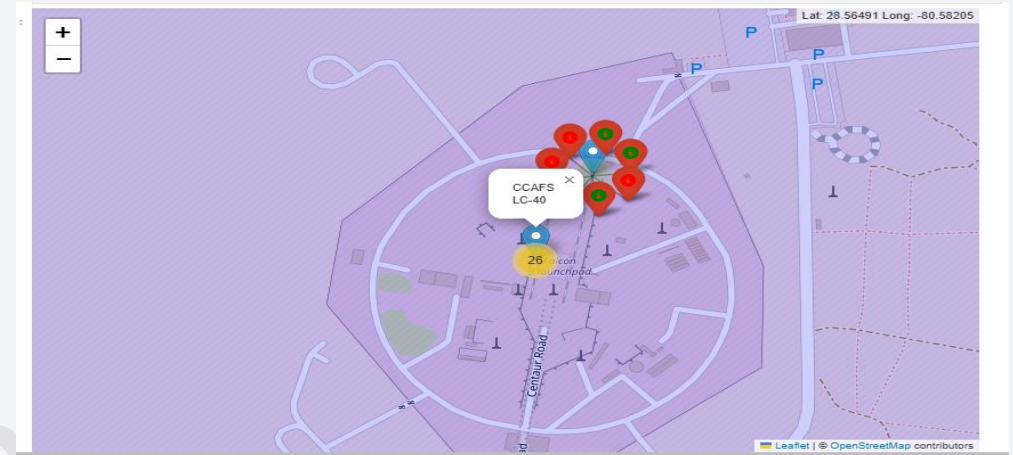
[https://github.com/ibrahimlois58-sketch/project-1/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20\(2\).ipynb](https://github.com/ibrahimlois58-sketch/project-1/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20(2).ipynb)

Build an Interactive Map with Folium

Folium map built using specific lat. and lon. and map objects such as `folium maker`, `folium.circle`, `folium.line` were added. This is to mark out the exact location of launch sites, identify successful and failed launches and also trace nearby railway, highway and city and the distance from launch sites to coastline

GitHub URL:

[https://github.com/ibrahimlois58-sketch/project-1/blob/main/lab_jupyter_launch_site_location%20\(2\).ipynb](https://github.com/ibrahimlois58-sketch/project-1/blob/main/lab_jupyter_launch_site_location%20(2).ipynb)



Build a Dashboard with Plotly Dash

Plots used

- Pie Chart
- Scatter plot

Interactions:

- Built-In Dropdown Trigger used on Pie chart versus losses (O) for that isolated location.
- Built-In Dual-Input Dependency used in Scatter plot This plot reacts to both controls simultaneously.

Why This Plot & Interaction?

- It makes it easy for Executives and stakeholders to view raw spreadsheets to see if a launch site is performing well. A pie chart is the fastest way to convey a part-to-whole ratio. It shows precise percentage of wins (1)

GitHub URL:

<https://github.com/ibrahimlois58-sketch/project-1/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)

Summary of how I built, evaluated, improved, and found the best performing classification model

- *Data Preparation: Standardized feature data X using StandardScaler and split it into 80% training and 20% testing sets (18 test samples).*
- *Leveraged 10-fold Stratified Cross-Validation via GridSearchCV to prevent overfitting and systematically optimize model configurations.*
- *Evaluated four distinct paradigms: Logistic Regression, Support Vector Machine (SVM), Decision Tree Classifier, and k-Nearest Neighbors (KNN).*
 - *Logistic Regression: Optimized regularization strength (C).*
 - *SVM: Tested linear, radial basis function (RBF), polynomial, and sigmoid kernels along with cost (C) and gamma scale variations.*
 - *Decision Tree: Sifted through criteria variations, tree depths, maximum feature limits, and leaf splitting constraints.*
 - *KNN: Altered nearest neighbor boundaries (1 through 10), internal algorithms, and Minkowski power metrics (p).*

Predictive Analysis (Classification)

- Error Diagnostic Evaluation: Utilized Confusion Matrices alongside validation test sets to pinpoint vulnerabilities, determining that the principal model weakness was False Positives (predicting a safe landing when it crashed).

Performance:

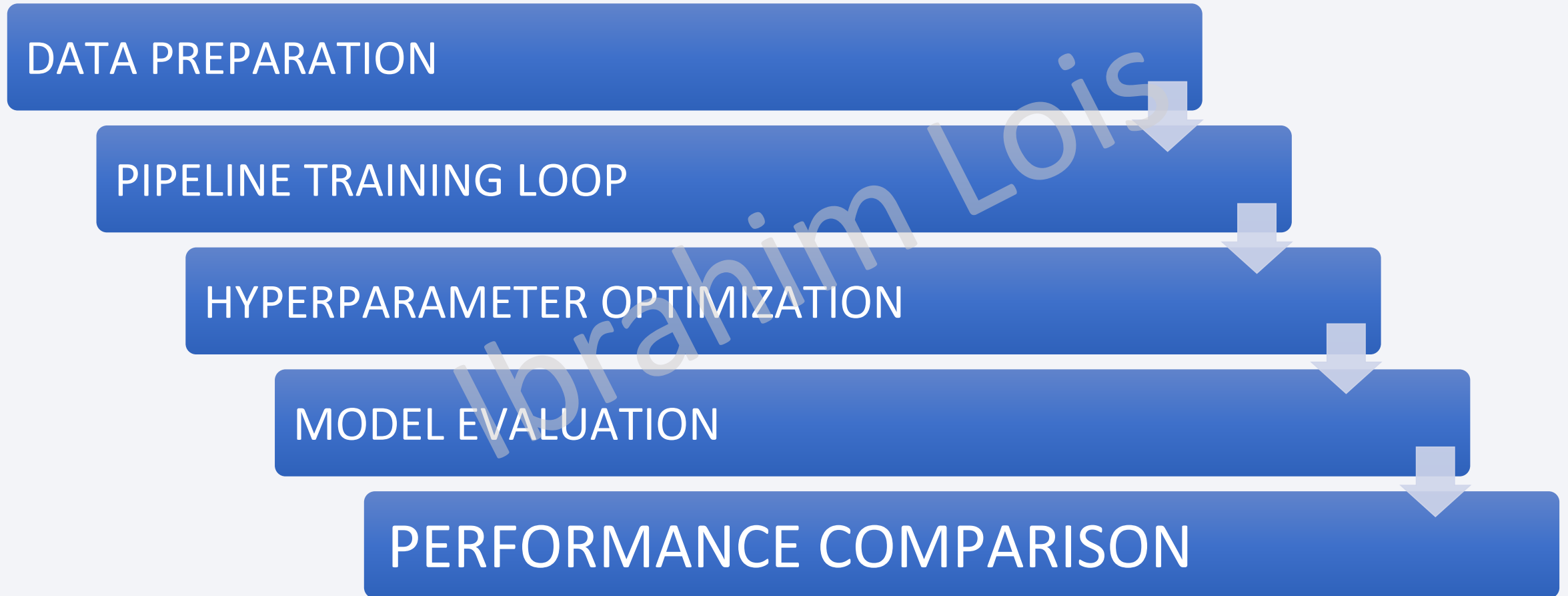
- All optimized variations converged successfully, producing identical validation test score realities of 83.33\% operational accuracy across the board.
- You need present your model development process using key phrases and flowchart

GitHub URL:

[https://github.com/ibrahimlois58-sketch/project-1/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20\(1\).ipynb](https://github.com/ibrahimlois58-sketch/project-1/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb)

Predictive Analysis (Classification)flowchart

Flowchart showing the structured path taken to go from raw datasets to final baseline comparison.



Results: Exploratory data analysis results

Unique Launch Sites where space missions took place

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

Total Payload Mass for NASA(CRS)

- 45596kg

Average Payload Mass for Booster Version 'F9 v1.1'

- 2928.40 kg

First Successful Landing Date on a Ground

Pad

- 2015-12-22

Boosters with Successful Drone Ship Landings and Payloads Between 4,000 and 6,000 kg

- F9 FT B1020
- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Exploratory data analysis results

Total Successful vs Failed Mission Outcomes

- Total successful mission = 98
- Total failed missions = 0

Booster Versions with maximum Payload Mass

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1048.5
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2

• F9 B5 B1058.3

• F9 B5 B1051.6

• F9 B5 B1060.3

• F9 B5 B1049.7

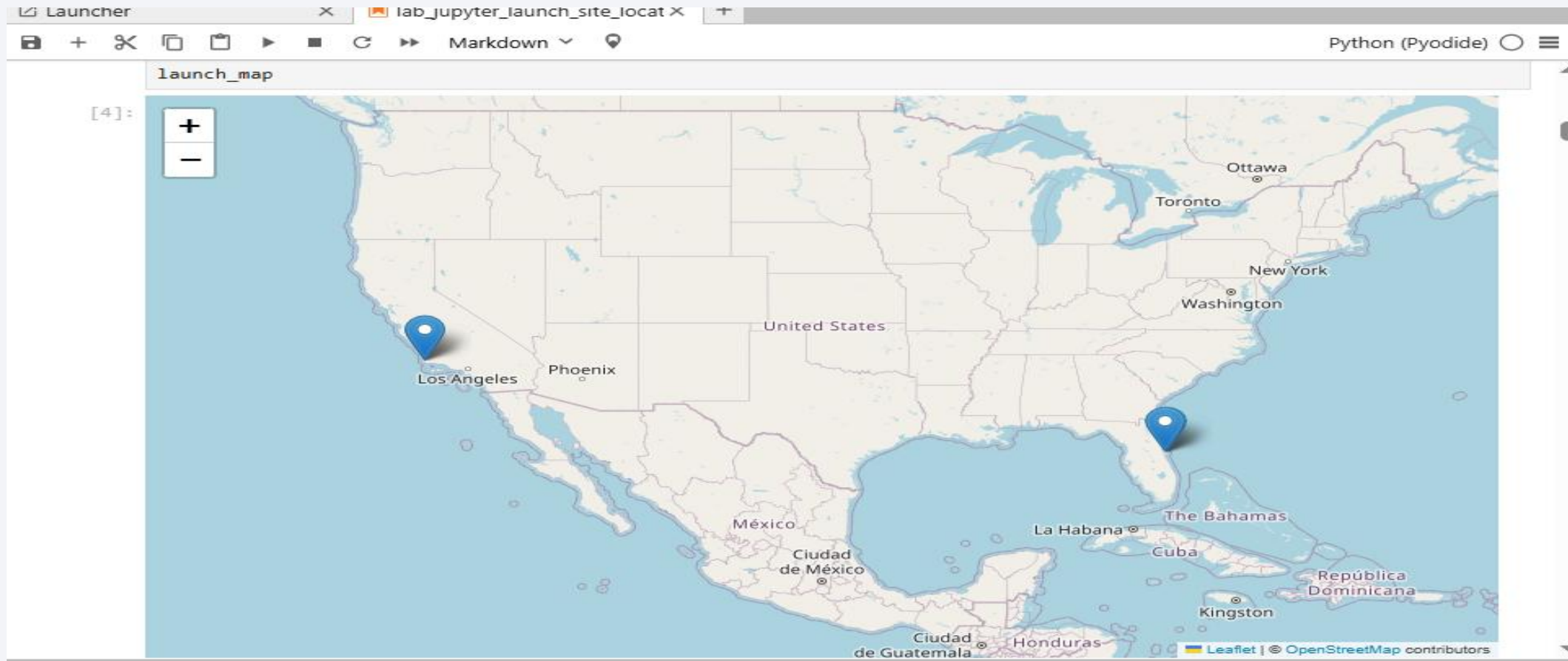
• 2015 Failed Drone Ship Landings

• January: F9 v1.1 B1012, CCAFS LC-40

• April: F9 v1.1 B1015, CCAFS LC-40

Interactive analytics demo in screenshots

All Launch Sites



Interactive analytics demo in screenshots Launch sites, Success vs Failed Launches

Launch sites, Success vs Failed Launches



Predictive analysis results

Algorithm Tuning and Performance Breakdown

Logistic Regression

- Tuned Hyperparameters: {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
- Training Cross-Validation Accuracy: 84.64%
- Test Data Accuracy: 83.33%
- Confusion Matrix Insights: The model distinguished between the classes giving 3 False Positives, 12 True Positives, 3 True Negatives, and 0 False Negatives.

Support Vector Machine (SVM)

- Tuned Hyperparameters: {'C': 1.0, 'gamma': 0.0316, 'kernel': 'sigmoid'}
- Training Cross-Validation Accuracy: 84.82%
- Test Data Accuracy: 83.33%
- Confusion Matrix Insights: Produced the exact same confusion matrix structure as Logistic Regression.

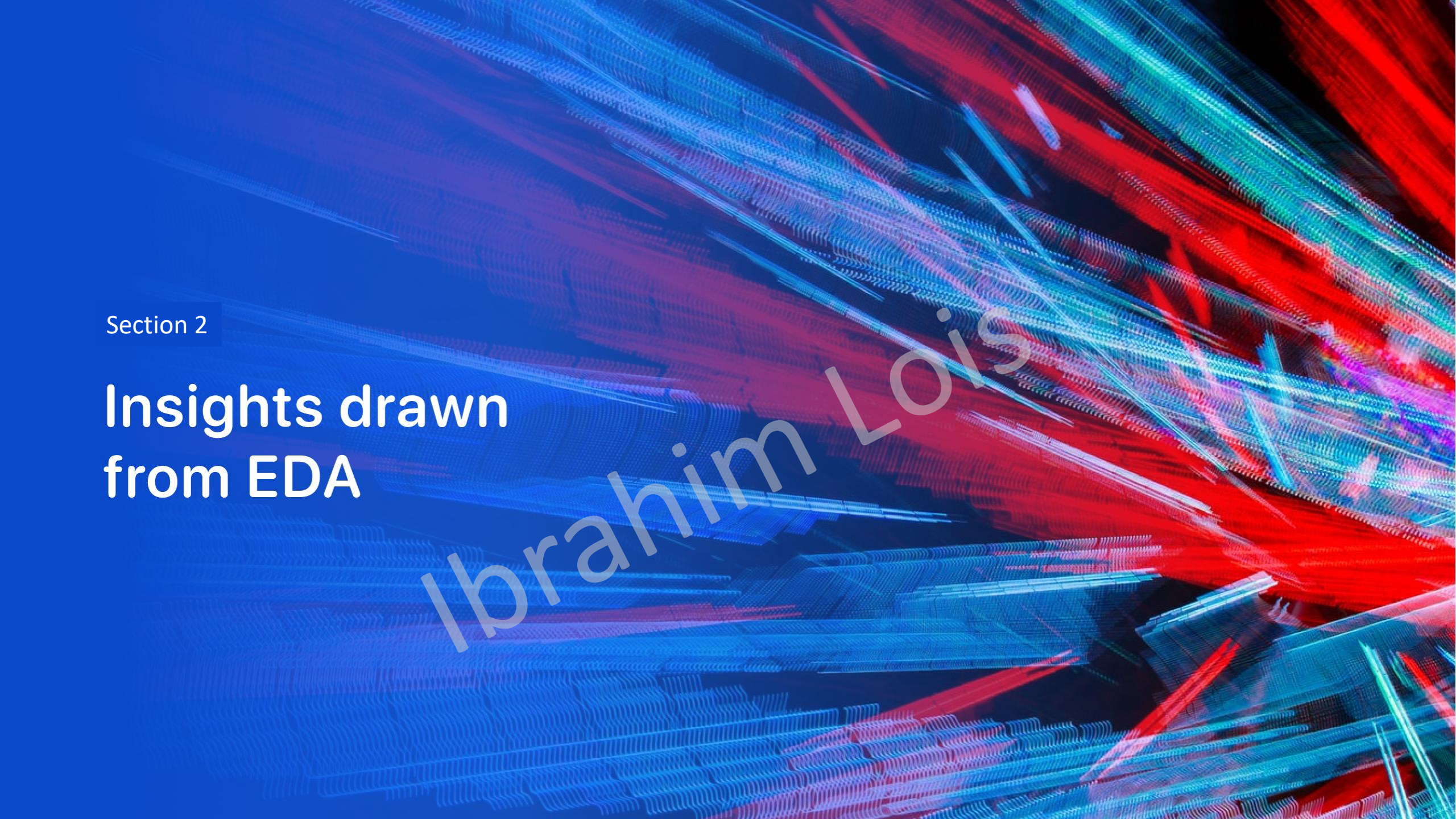
Predictive analysis results

Decision Tree Classifier

- Tuned Hyperparameters: {'criterion': 'gini', 'max_depth': 12, 'max_features': 'log2', 'min_samples_leaf': 4, 'min_samples_split': 10, 'splitter': 'best'}
- Training Cross-Validation Accuracy: 87.50%
- Test Data Accuracy: 83.33%
- Confusion Matrix Insights: Maintained identical testing performance to the previous models, experiencing the exact same 3 False Positive errors on the test set.

k-Nearest Neighbors (KNN)

- Tuned Hyperparameters: {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}
- Training Cross-Validation Accuracy: 84.82%
- Test Data Accuracy: 83.33%
- Confusion Matrix Insights: It generated the same test matrix breakdown.

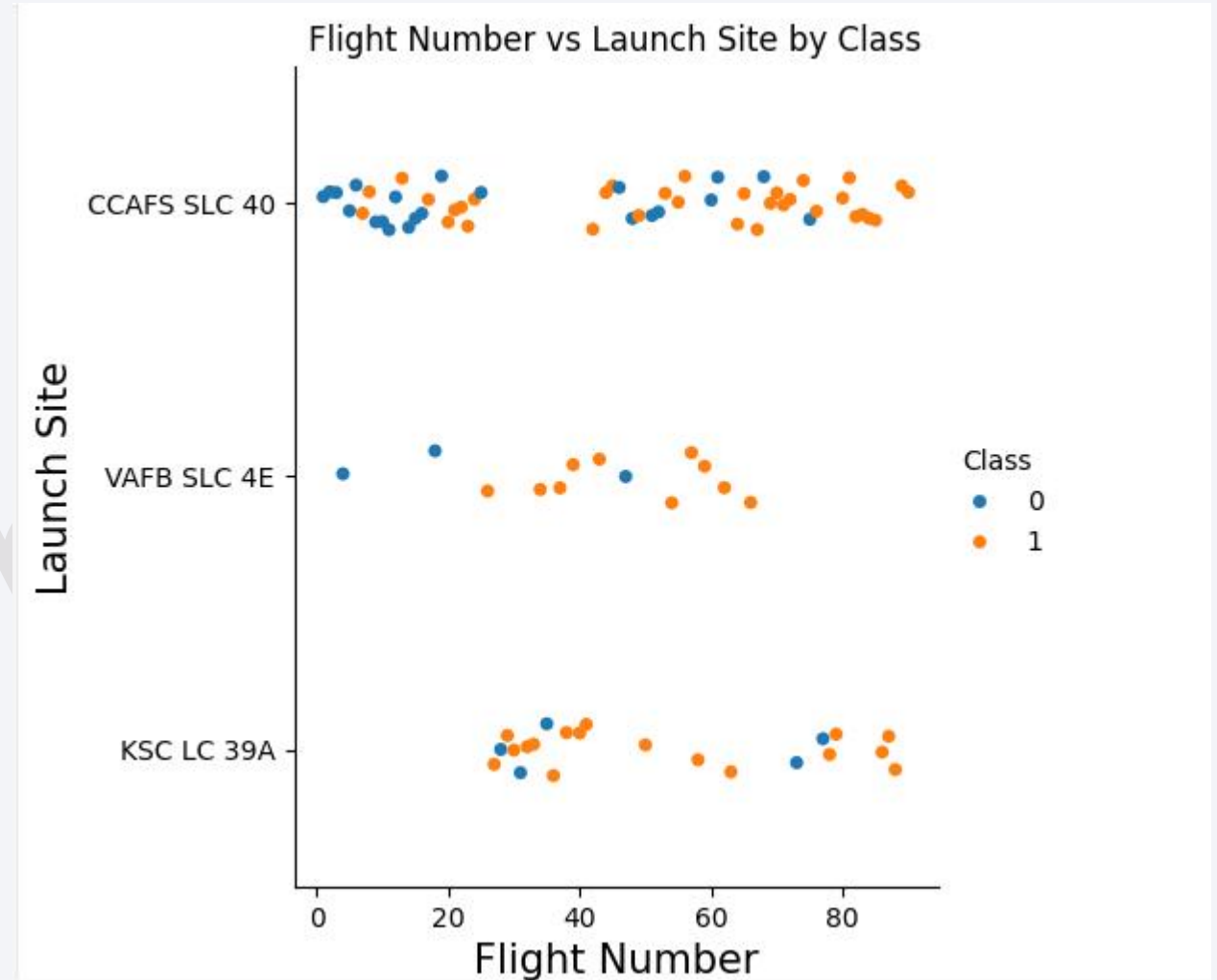


Section 2

Insights drawn from EDA

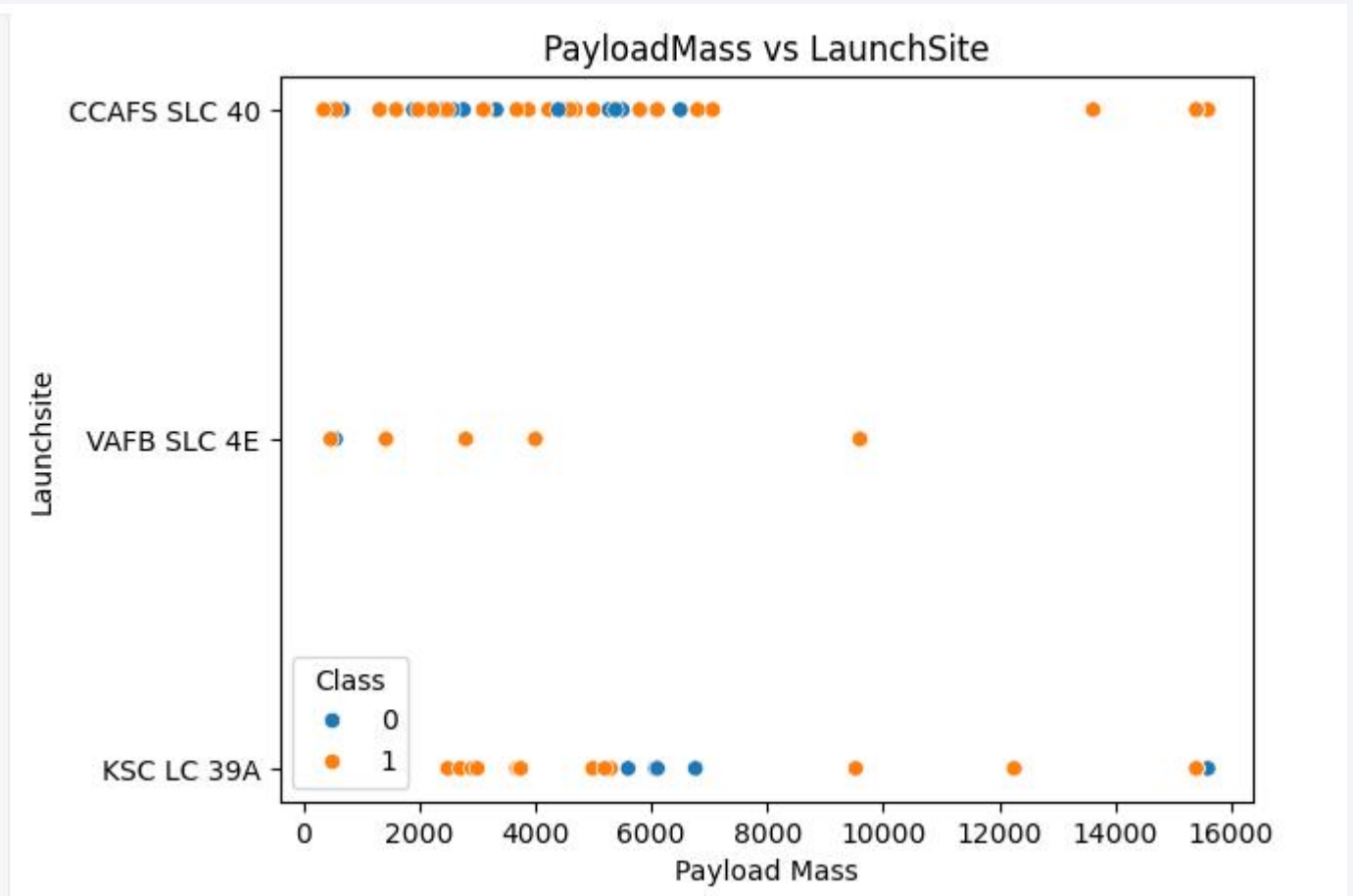
Flight Number vs. Launch Site

- **VAFB SLC 4E** has fewer launches at the beginning, more launches after 25 with increasing success rate and 0 launch after 70.
- **KSC LC 39A** has not launch until after 25. a few failures but more successes.
- For **CCAFS SCL 40**, there is no distinct failed or successful launch. Bth are mixed all together showin a contineus fail and success rate launches with a break between 20 - 40.



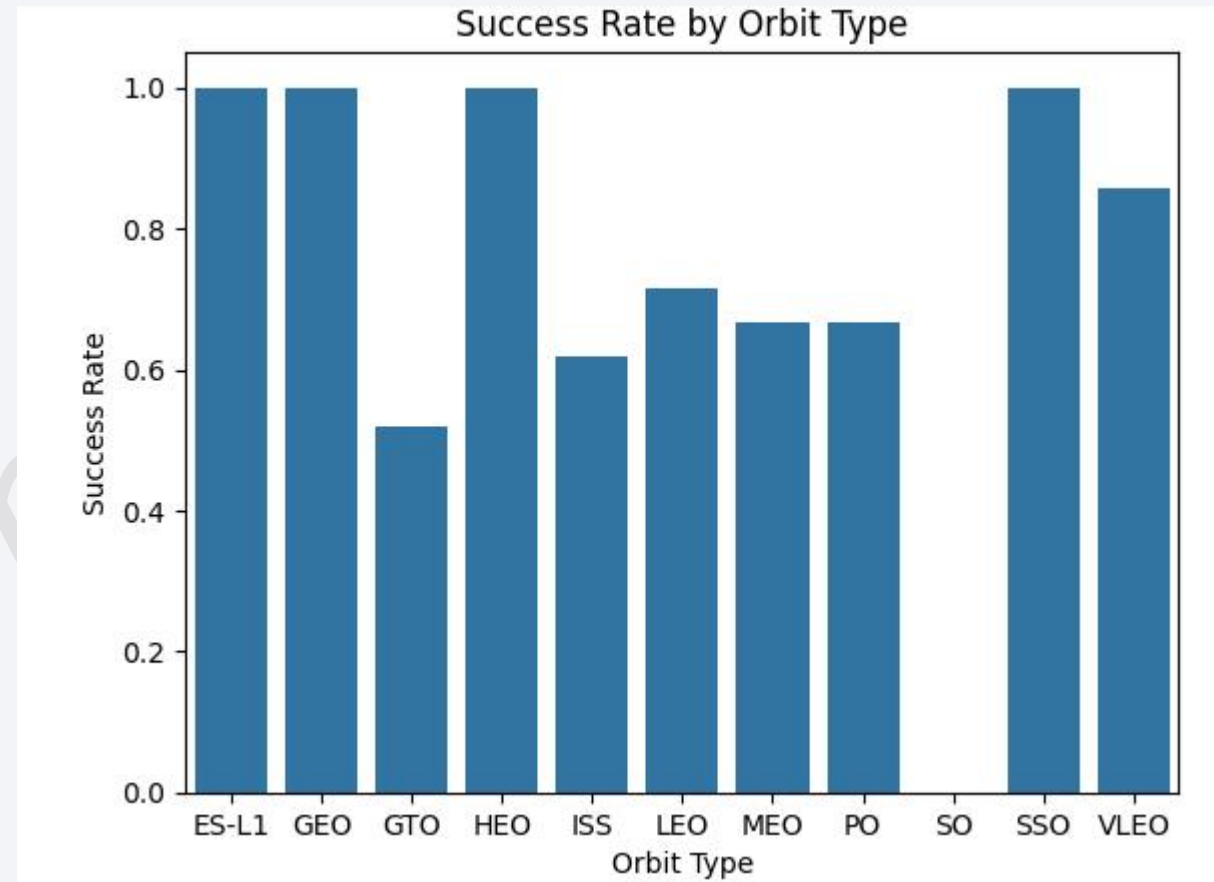
PayloadMass vs. Launch Site

- Payload Mass Vs. Launch Site scatter point chart shows **VAFB-SLC** launchsite has no rockets launched for heavypayload mass(greater than 10000).
- **CCAFS SLC 40** has the highest success rocket launched with rocket launched at as high as 16000 heavy payload



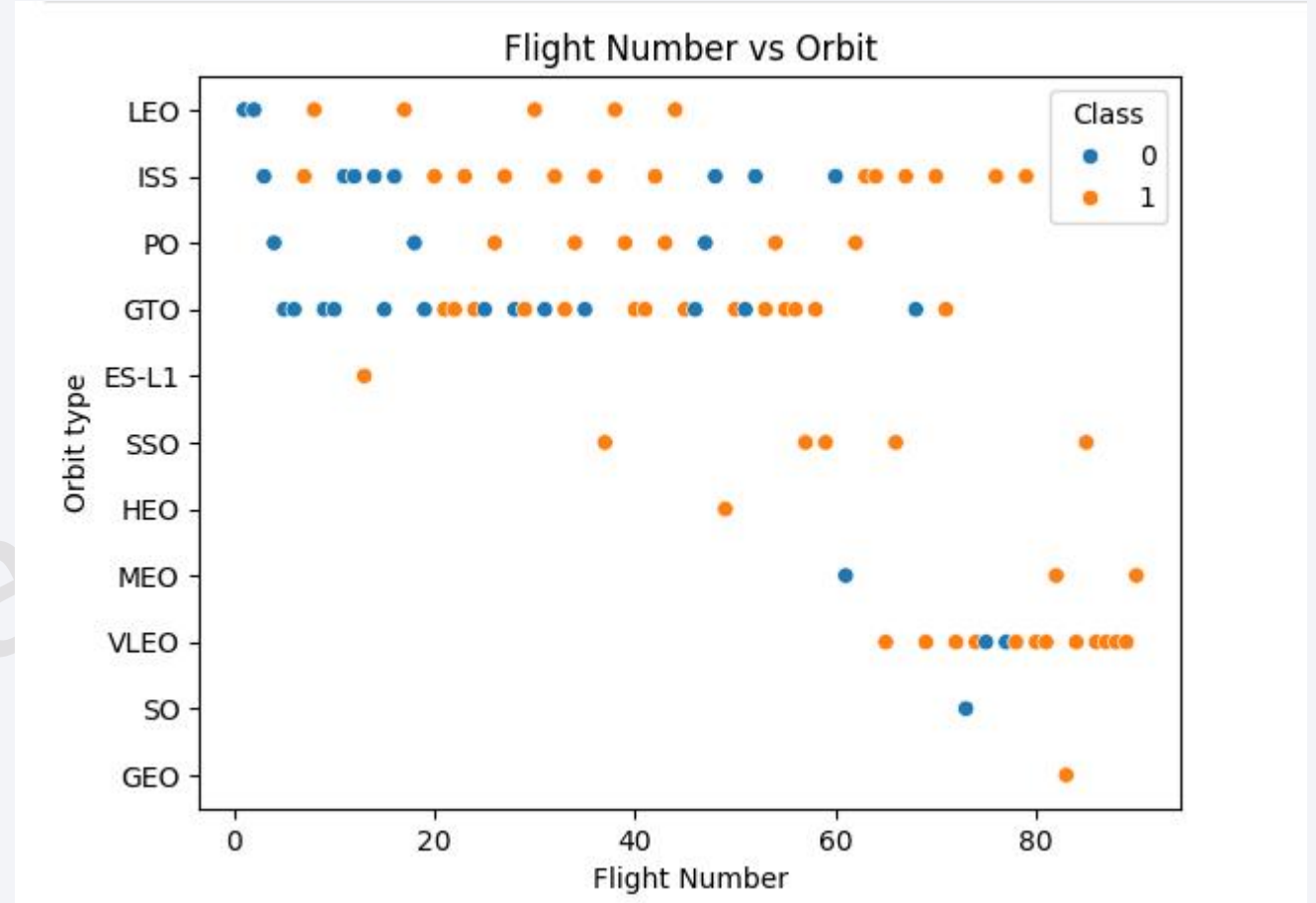
Success Rate vs. Orbit Type

The bar chart shows that *SSO*, *HEO*, *GEO* and *ES-L1* orbits have the highest success rates whereas *GTO* have the lowest. *SO* has 0 success launch.



Flight Number vs. Orbit Type

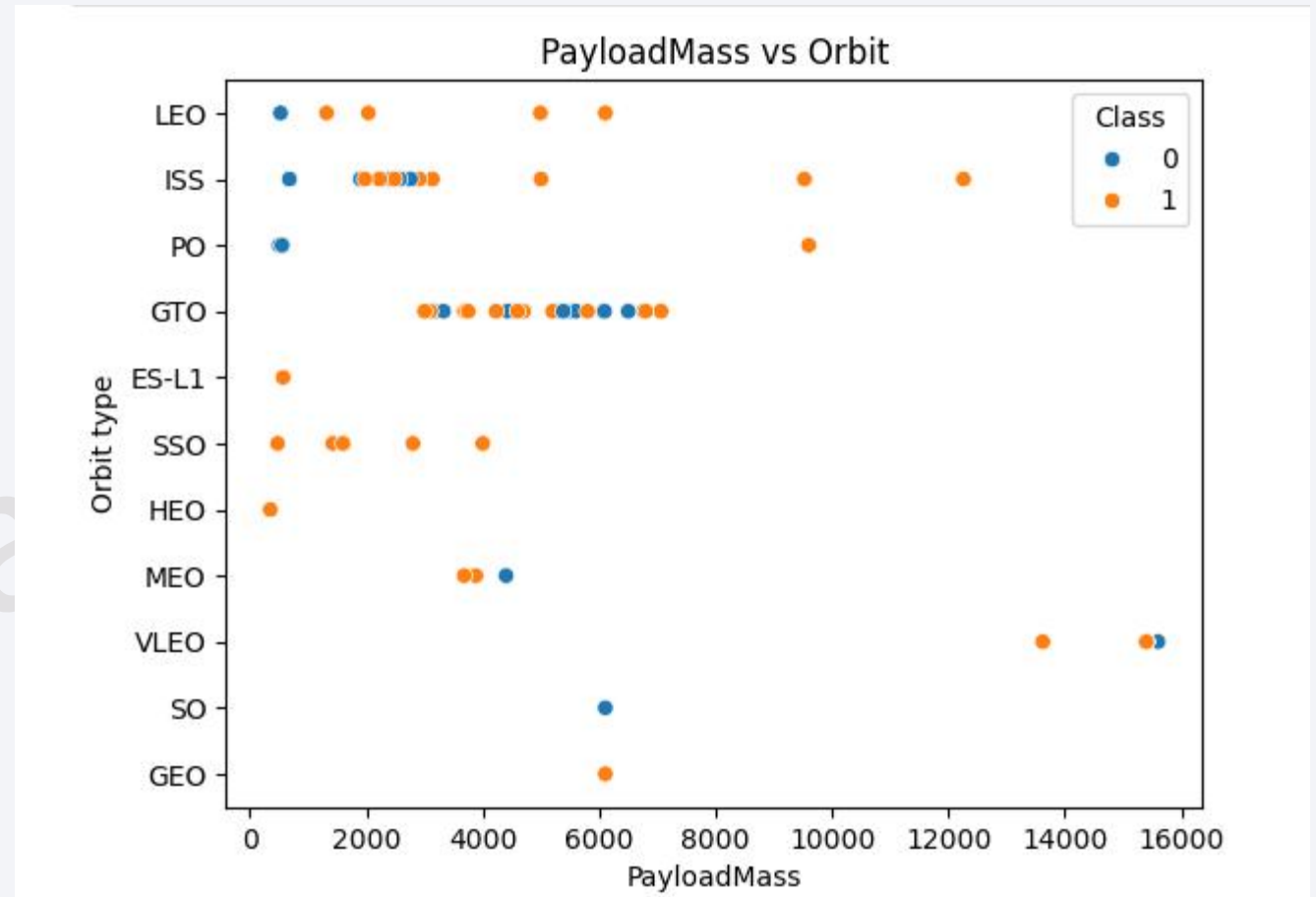
Observe that in the *LEO* orbit, success seems to be related to the number of flights. Conversely, in the *GTO* orbit, there appears to be no relationship between flight number and success.



PayloadMass vs. Orbit Type

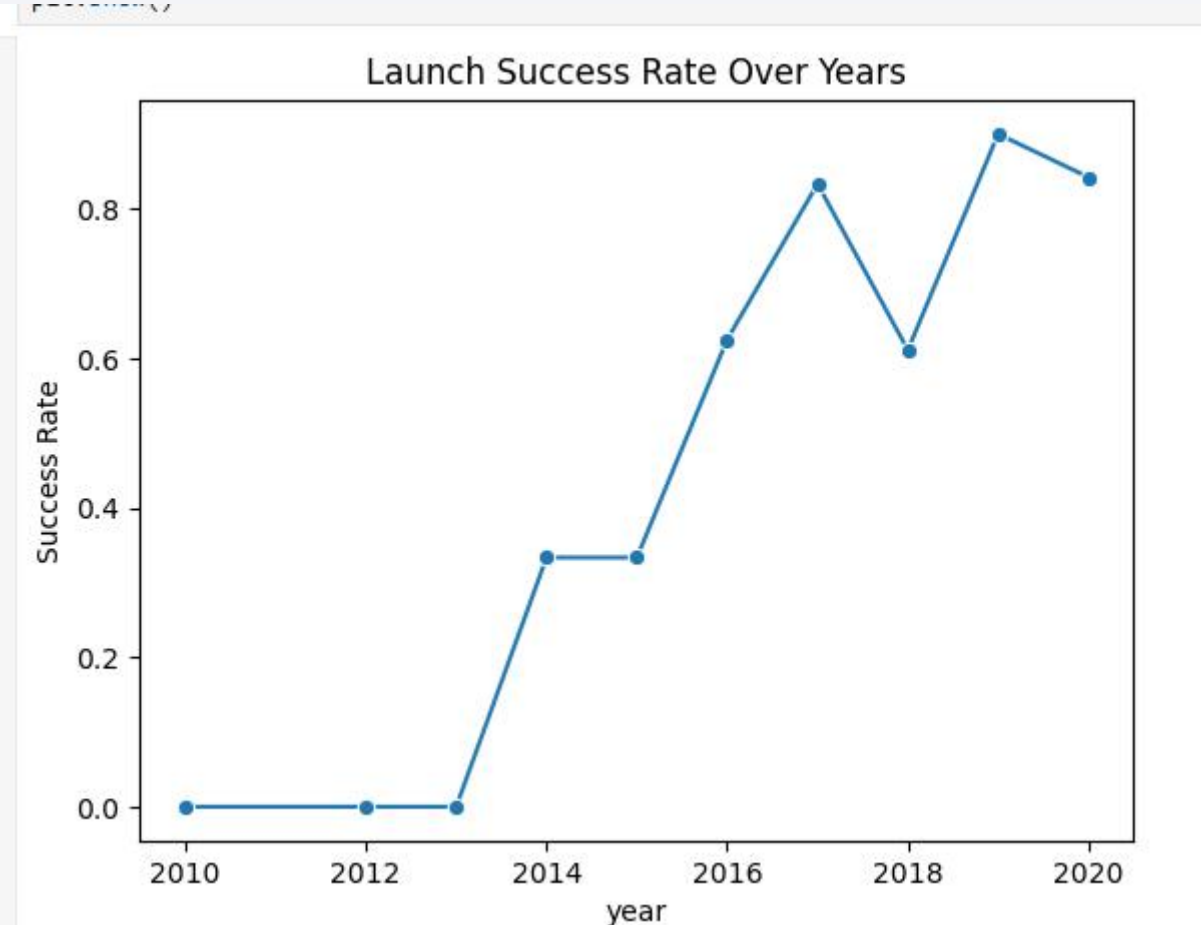
- With heavy payloads the successful landing or positive landing rate are more for *PO, LEO and ISS*.

However, for *GTO*, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend

Observe that the success rate since 2013 kept increasing till 2020



All Launch Site Names

Below are the launch site names used in analysis ;

- CCAFS LC-40
- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

These Launch sites in themselves cannot be used without their Latitude and Longitude.

Launch Site Names that Begin with 'CCA'

- Lat and Long represents Latitude and Longitude respectively
- Class represents successful or failed launch with 0 showing failed launches all through the 5 launch sites
- Maker Color showing red for failed (0) class

	Launch Site	Lat	Long	class	maker color
0	CCAFS LC-40	28.562302	-80.577356	0	red
1	CCAFS LC-40	28.562302	-80.577356	0	red
2	CCAFS LC-40	28.562302	-80.577356	0	red
3	CCAFS LC-40	28.562302	-80.577356	0	red
4	CCAFS LC-40	28.562302	-80.577356	0	red

Total Payload Mass

The total payload carried by boosters from NASA(CRS) = 45596

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Average Payload Mass by F9 v1.1

Average payload mass carried by booster version F9 v1.1:
2928.40 kg

- This shows where SpaceX started hitting their stride commercially, giving a baseline to appreciate how much they increased the rocket's power and capacity in just a few years.

First Successful Ground Landing Date

The first Falcon 9 rocket that successfully landed on a ground pad landed on December 22, 2015 with the serial number B1019.

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Successful Drone Ship Landing with Payload between 4000 and 6000

Boosters with successful drone ship and payload mass between 4000 and 6000:

- F9 FT B1020
- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Out of all SpaceX launches, these specific missions were the only ones that carried a medium-heavy payload (weighing between 4,000 and 6,000 kg) AND successfully landed the first-stage rocket on an ocean drone ship. This heavy-lifting rockets that proved SpaceX could successfully land and reuse boosters even when carrying massive payloads.

Total Number of Successful and Failure Mission Outcomes

The total number of successful and failure mission outcomes

- Total successful missions: 98
- Total failed missions: 0

This simply says there are no failed landing mission outcomes

Boosters Carried Maximum Payload

Names of the booster which have carried the maximum payload mass include;

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1048.5
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2
- F9 B5 B1058.3
- F9 B5 B1051.6
- F9 B5 B1060.3
- F9 B5 B1049.7

2015 Launch Records

In January and April 2015, there was had List the failed landing_outcomes in drone ship. it is shown below along side their booster versions, and launch site names.

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

This shows the count of landing outcomes (such as Failure (drone ship), Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

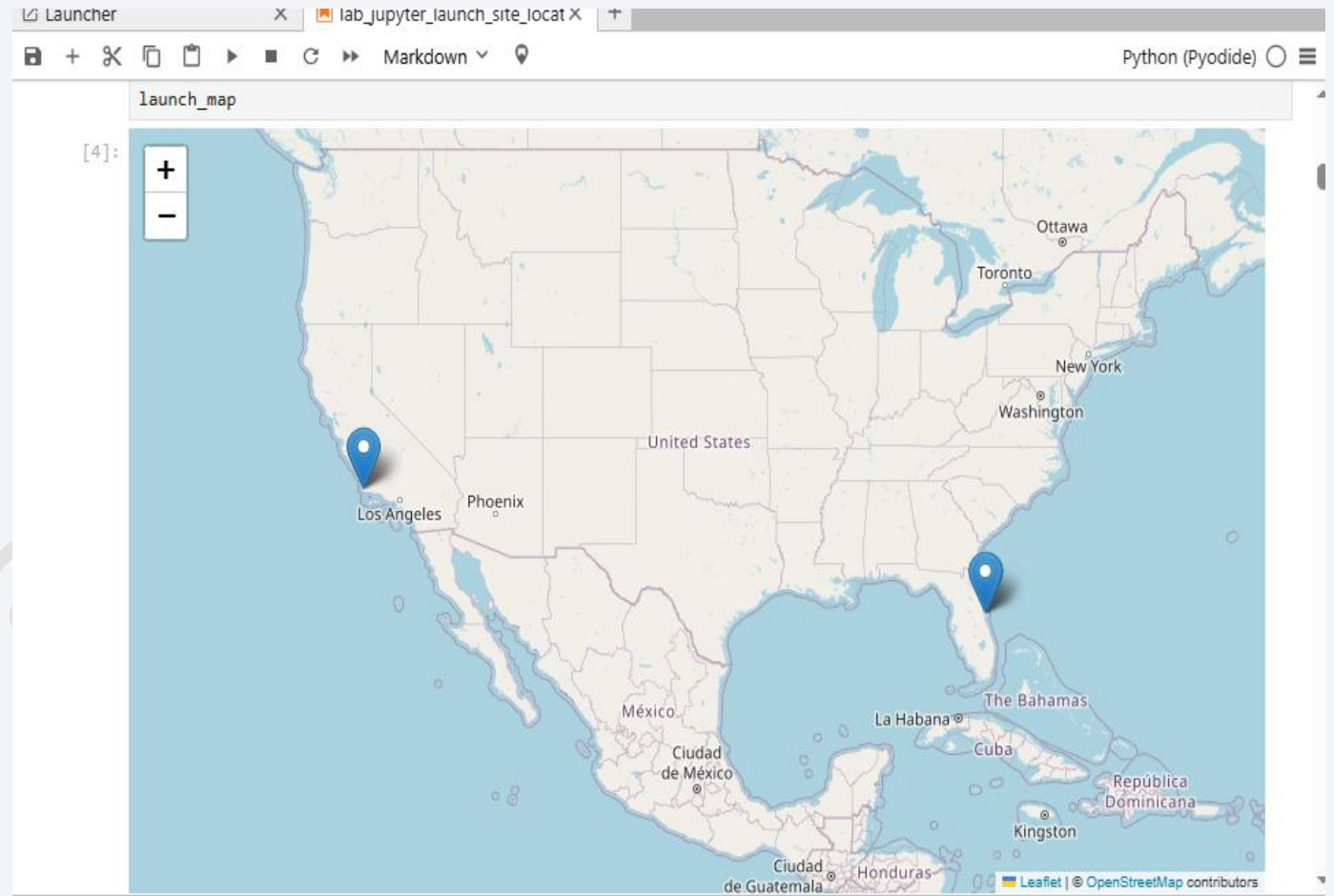
Section 3

Launch Sites Proximities Analysis

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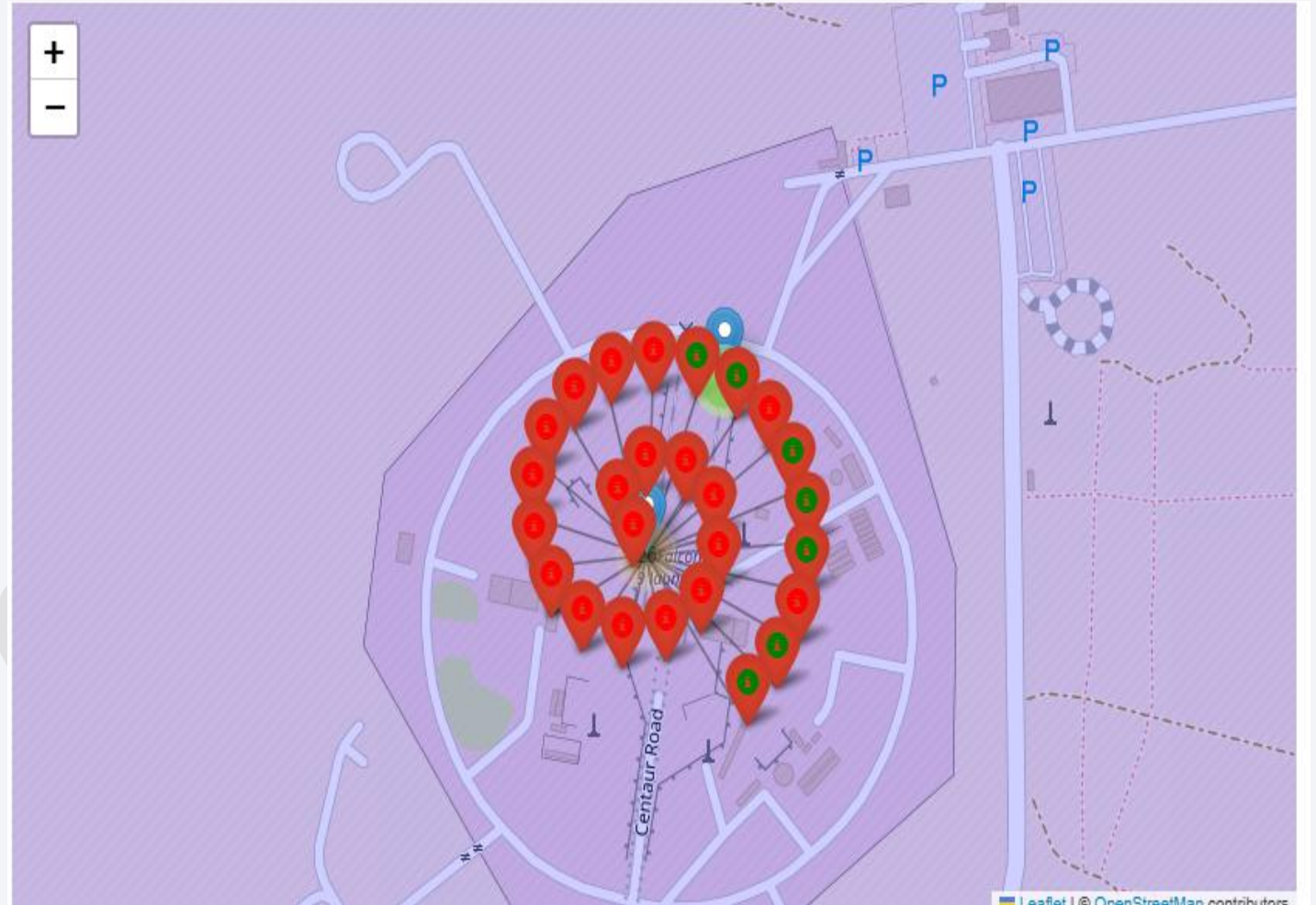
Global Map of All Launch Sites

All launch site; *KSC LC-39A*, *CCAFS SLC-40*, *CCAFS LC-40*, *VAFB SLC-4E* are located on the map. this might only show two locations but zooming on the interactive map expands giving all locatuons seperately.



Landing Outcomes of CCAFS LC-40 Launch Site

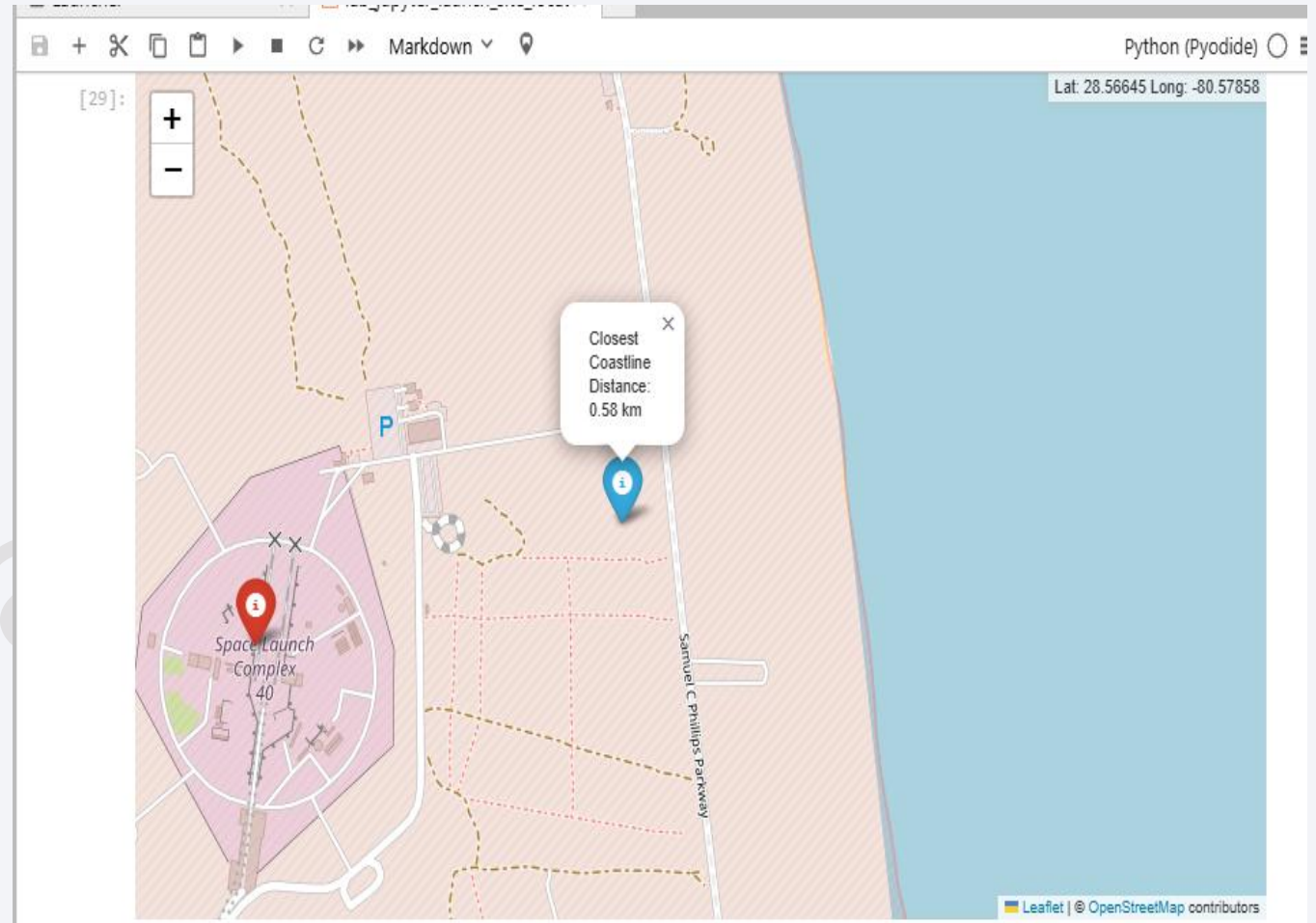
This map shows the color-labeled launch outcomes of the launch site **CCAFS LC-40** with 7 successful launches shown in green color and 19 failed launches shown in red color. This site has more failed than successful rocket launches.



Coastline Distance from a Launch Site

This map shows a launch site with the closest proximity of 0.58km to the closest coastline.

- The red location marker shows the launch site location, blue marker shows the coastline.



Section 4

Build a Dashboard with Plotly Dash

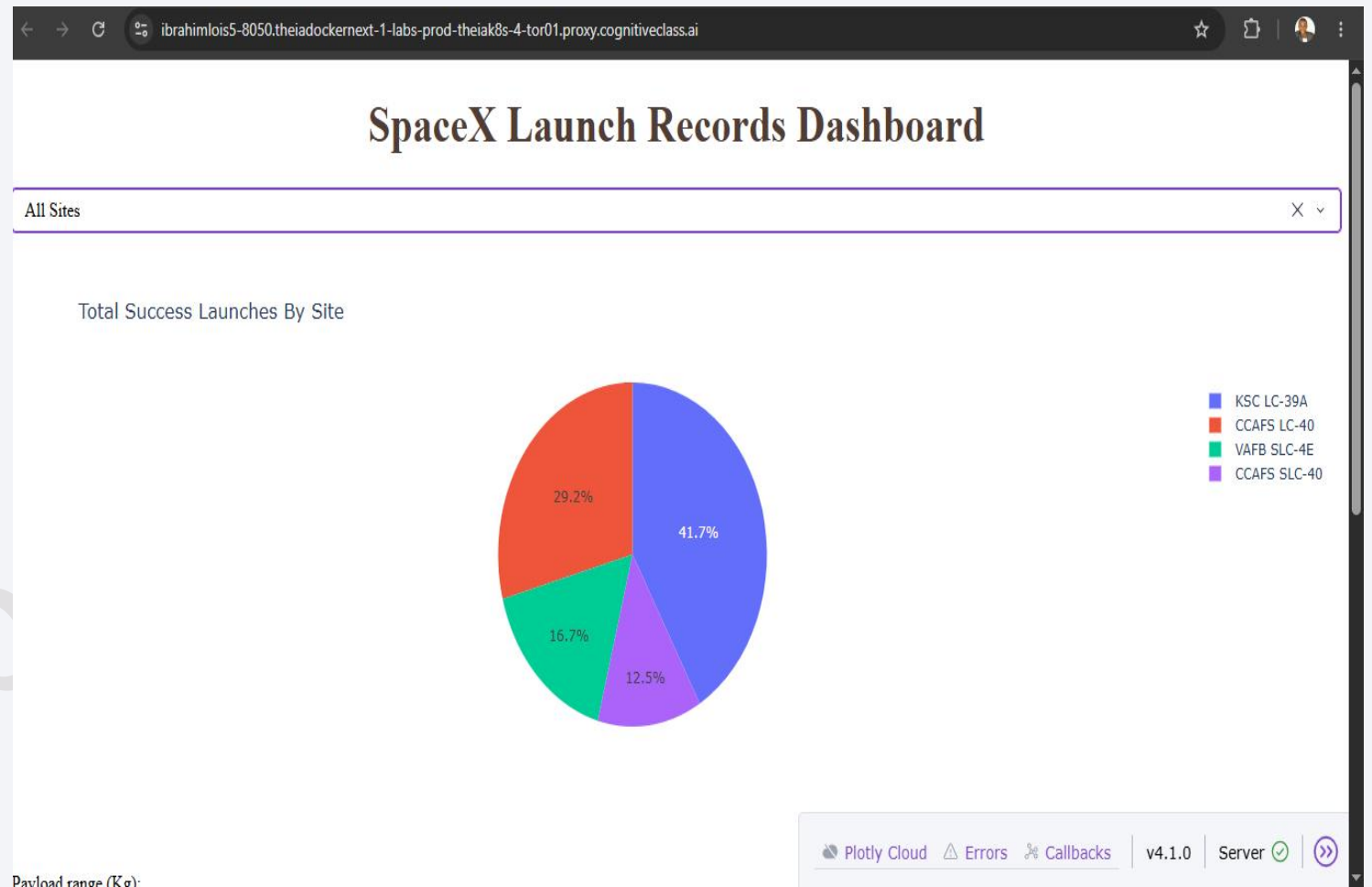
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Launch Success Count For All Sites

A Piechart showing the Success count for all site by percent in decending order;

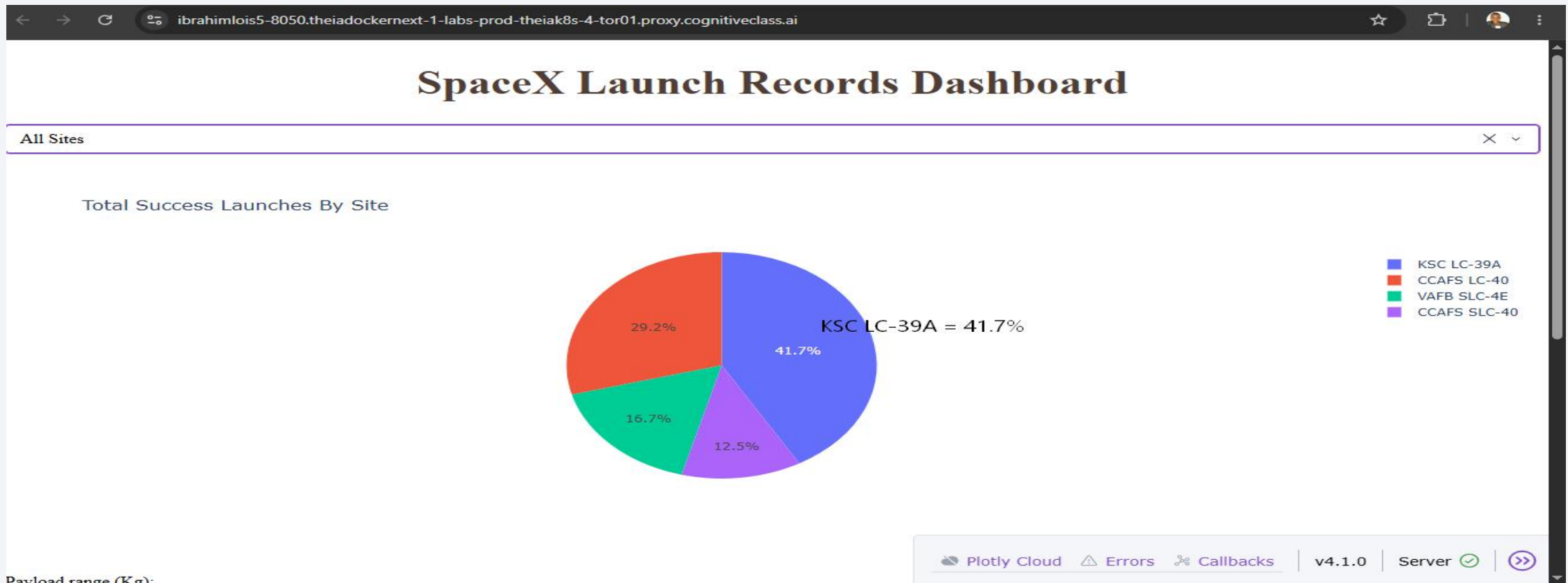
- KSC LC-39A 41.7%
- CCAFS LC-40 29.2%
- VAFB SLC-4E 16.7%
- CCAFS SLC-40 12.5%

From this success count, KSC LC-39A has the highest success lauch and CCAFS SLC-40 has the lowest success count.



Highest Launch Success screenshot

Piechart showing the highest success launch which is **KSC LC-39A** with launch success of **41.7%**.



Payload range (Kg):

Payload vs. Launch Outcome scatter plot for all sites

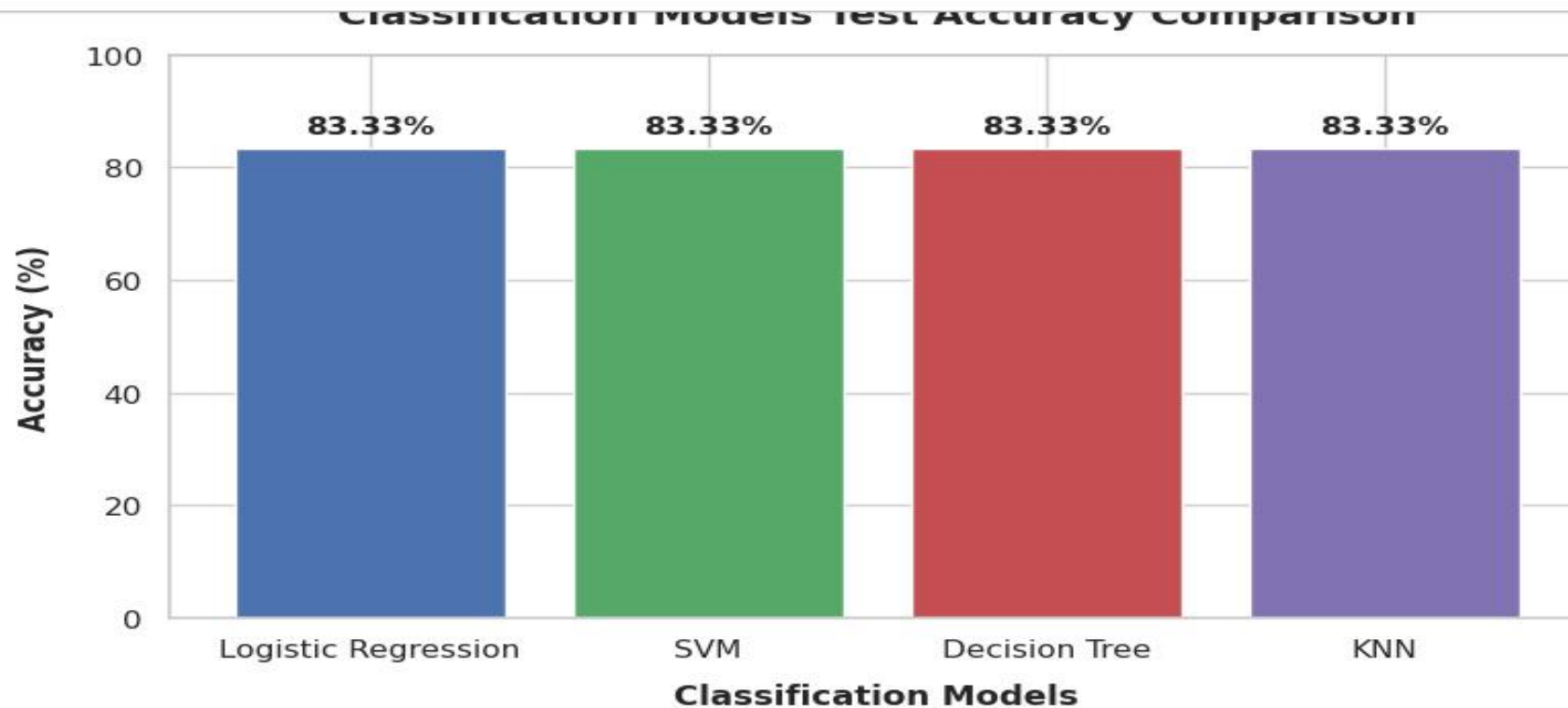
screenshot of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider



Classification Accuracy

Built model accuracy for all built classification models in a bar chart

- All models have the same classification accuracy of 83.33%



Section 5

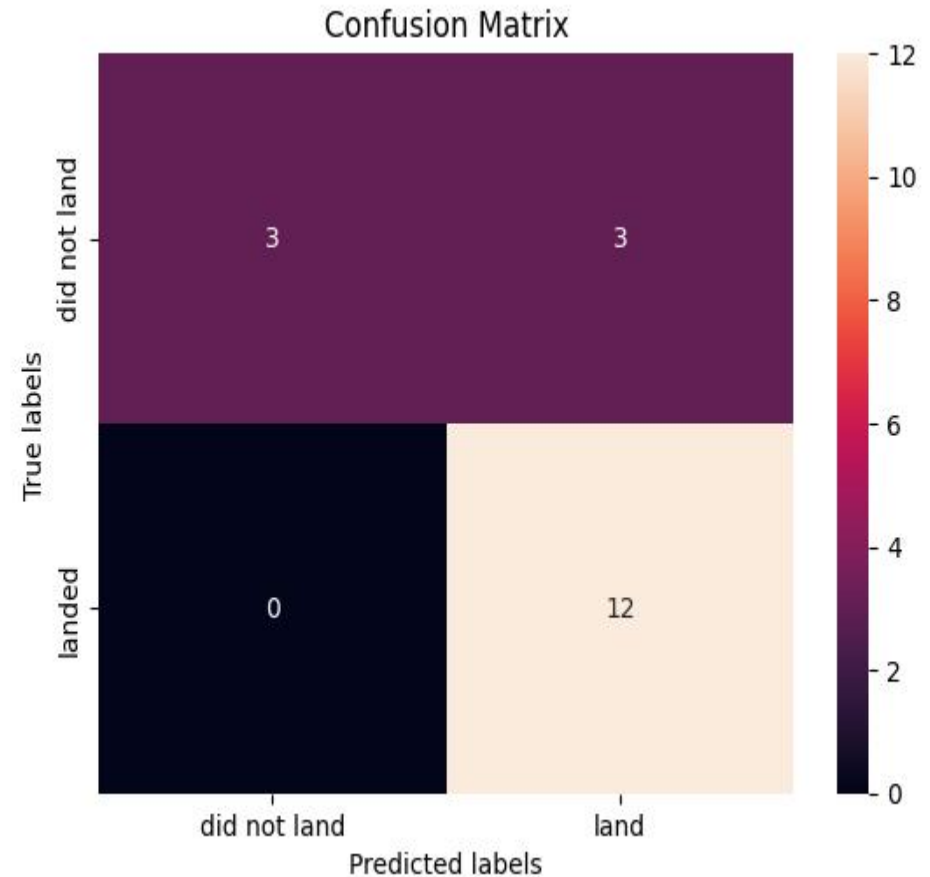
Predictive Analysis (Classification)

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Confusion Matrix

Since all models have the same accuracy, the matrix hence is same for all four classification models—Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN). True Positives (12): The model correctly predicted that the first stage would land.

- True Negatives (3): The model correctly predicted that the first stage would not land.
- False Positives (3): The model incorrectly predicted a successful landing when it actually failed.
- False Negatives (0): The model perfectly identified every successful landing without missing any.



Conclusions

- The first Falcon 9 rocket that successfully landed on a ground pad landed on December 22, 2015 with the serial number B1019.
- Total Payload Mass for NASA(CRS) = 45596kg
- Average Payload Mass for Booster Version 'F9 v1.1' = 2928.40 kg
- All models perform equally well with an accuracy of 83.33%
- The bar chart show that **SSO**, **HEO**, **GEO** and **ES-L1** orbits have the highest success rates whereas **GTO** have the lowest. **SO** has 0 success launch.

Appendix

Github url for spacex-dash-app.py:

<https://github.com/ibrahimlois58-sketch/project-1/blob/main/spacex-dash-app.py>

Github url for spacex_launch_dash.csv:

https://github.com/ibrahimlois58-sketch/project-1/blob/main/spacex_launch_dash.csv

Thank you!

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